

# Jarrett D. Phillips, Ph.D.

School of Computer Science  
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GitHub: [jphill01](#)

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## RESEARCH AREAS

Computational statistics · Data science · Machine learning · Statistical modelling · Biodiversity informatics · eDNA and DNA barcoding

## ACADEMIC APPOINTMENTS

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|-------------------------------------------------------------------------------|--------------|
| <b>Adjunct Professor</b> , School of Computer Science<br>University of Guelph | 2023–Present |
| <b>Affiliated Faculty</b> , One Health Institute<br>University of Guelph      | 2026         |

## EDUCATION

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| <b>Ph.D. in Computational Sciences</b> , University of Guelph<br>Co-Advisors: Dr. Daniel Gillis and Dr. Robert Hanner<br>Advisory Committee Members: Dr. Deborah Stacey and Dr. Graham Taylor<br>Dissertation: A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling Completeness: Implications for DNA Barcode Gap Estimation | 2022 |
| <b>Master of Bioinformatics</b> , University of Guelph<br>Co-Advisors: Dr. Robert Hanner and Dr. Daniel Ashlock<br>Major Paper: Assessing DNA Barcode Haplotype Sampling Diversity in the Ray-finned Fishes (Chordata: Actinopterygii)                                                                                                                    | 2014 |
| <b>BSc. (Hons.)</b> in Biological Science, University of Guelph<br>Coursework in bioinformatics, ecology, evolutionary biology, comparative animal physiology, genetics, mathematics, and statistics                                                                                                                                                      | 2013 |

## RESEARCH EXPERIENCE

|                                                                                                                                                          |              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Postdoctoral Fellow</b><br>University of Guelph<br>GBADs Informatics Team<br>Stacey Lab, School of Computer Science<br>Supervisor: Dr. Deborah Stacey | 2023–Present |
| <b>Postdoctoral Fellow</b><br>University of Guelph<br>Gillis Lab, School of Computer Science                                                             | 2023–Present |

Hanner Lab, Department of Integrative Biology  
Supervisors: Drs. Daniel Gillis and Robert Hanner

**Postdoctoral Fellow**

2022

University of Guelph

Hanner Lab, Department of Integrative Biology

Supervisor: Dr. Robert Hanner

## **RESEARCH FUNDING**

### Completed

8. **Food From Thought Advancing Research Impact (ARIF) Fund – Livestock Innovation Grant** 2024  
University of Guelph \$40,000 CAD  
A Modular Decision- and Policy-Making Tool for Global Livestock Disease Burden Assessment  
**Role:** Co-applicant
7. **Food From Thought Advancing Research Impact (ARIF) Fund** 2023  
University of Guelph \$30,000 CAD  
Forecasting Seafood Fraud Occurrence in the Supply Chain Within Major Canadian Cities Using Bayesian Hierarchical Binary Logistic Time-Series Regression Modeling  
**Role:** HQP
6. **SoCS Travel Grant** 2019  
University of Guelph \$1,000 CAD
5. **Arthur D. Latornell Graduate Travel Grant** 2019  
University of Guelph \$500 CAD
4. **Graduate Teaching Assistantships** 2016–2020  
University of Guelph \$35,000 CAD
3. **Graduate Research Assistantships** 2017–2019  
University of Guelph \$11,000 CAD
2. **CPES Graduate Dean's Scholarship** 2017  
University of Guelph \$3,500 CAD
1. **CPES Graduate Excellence Entrance (GEE) Scholarship** 2016  
University of Guelph \$30,000 CAD

### Submitted

3. **NSERC Discovery Grant** 2026  
University of Guelph Amount \$TBD CAD  
Computational Statistical Methods for Specimen Sampling Sufficiency and Species Genetic Diversity Assessment  
**Role:** Principal Investigator
2. **NSERC Alliance Society Grant** 2026  
University of Guelph \$2.5 million CAD  
AI-Driven Biodiversity Restoration: Scalable Solutions for Ecosystem Monitoring and Conservation  
**Role:** Collaborator
1. **NSERC CREATE Grant** 2026  
University of Guelph \$9.9 million CAD  
AI-Driven Training of Next-Generation Conservation Scientists  
**Role:** Collaborator

### To be Submitted

1. **Advancing Research Impact Fund: “Quick Wins” Knowledge Mobilization Fund** 2026  
University of Guelph \$20,000 CAD  
GBADs ModelBuilder as a Tool to Teach Livestock Population Dynamics and Disease Burden Risk  
Estimation  
**Role:** Co-applicant

## **SOFTWARE AND COMPUTATIONAL TOOLS**

98,000+ software downloads through the Comprehensive R Archive Network (CRAN)

6. **Seafood Fraud Modelling App** · Python web application · URL: <https://dynamics-between-seafood-fraudsters-and-buyers.streamlit.app>
5. **GBADs ModelBuilder** · R Shiny web application · URL: <https://animalhealthmetrics.org/2025/10/14/introduction-to-gbads-new-model-builder-prototype/>
4. **RulesTools** · R package · >3000 downloads · DOI: [10.32614/CRAN.package.RulesTools](https://doi.org/10.32614/CRAN.package.RulesTools)
3. **Frequentist/Bayesian DNA barcode gap coalescent-informed estimators** · R scripts
2. **VLF (Very Low Frequency)** · R package · >40000 downloads · DOI: [10.32614/CRAN.package.VLF](https://doi.org/10.32614/CRAN.package.VLF)
1. **HACSim (Haplotype Accumulation Curve Simulator)** · R package · R Shiny web application · >50000 downloads · DOI: [10.32614/CRAN.package.HACSim](https://doi.org/10.32614/CRAN.package.HACSim)

## **STUDENT SUPERVISION AND MENTORSHIP**

Mentored/co-supervised **18 undergraduate and graduate students** across computer science, bioinformatics, statistics, and biology.

\* Indicates students under my direct mentorship/supervision

\*\* Indicates students under my co-mentorship/co-supervision

### Undergraduate

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|-----------------------------------------------------------------------------------------------------------|-----------|
| <b>*Brendan Carl Rosario</b>                                                                              | 2025      |
| Undergraduate Student Volunteer · Supervised Machine Learning for eDNA spatiotemporal sampling            |           |
| <b>*Richard Cui</b>                                                                                       | 2025      |
| Summer Undergraduate Research Assistant (URA) · Dynamical modelling of seafood fraud in the supply chain  |           |
| <b>*Nikolett Toth</b>                                                                                     | 2024–2025 |
| CIS*4900/4910 · Mining association rules for eDNA spatiotemporal sampling                                 |           |
| <b>*Nikolett Toth</b>                                                                                     | 2024      |
| Summer Undergraduate Research Assistant (URA) · Mining association rules for eDNA spatiotemporal sampling |           |
| <b>*Fynn De Vuono-Fraser</b>                                                                              | 2024      |
| CIS*4900/4910 · Bayesian modelling of seafood fraud in the Canadian supply chain                          |           |
| <b>*Zaid Al-Gayyali</b>                                                                                   | 2023      |
| Summer Undergraduate Research Assistant (URA) · Seafood Fraud Visualization Tool Shiny app                |           |
| <b>*Fynn De Vuono-Fraser</b>                                                                              | 2023      |
| STAT*4600 · Bayesian modelling of seafood fraud in the Canadian supply chain                              |           |
| <b>*Navdeep Singh</b>                                                                                     | 2021      |
| CIS*4900 · HACSim R Shiny web application                                                                 |           |
| <b>*Scarlett Bootsma</b>                                                                                  | 2020–2021 |
| CIS*4900/4910 · HACSim simulation study                                                                   |           |

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|------------------------------------------------------------------------------------------------|-----------|
| <b>**Maya Persram</b><br>Hanner Lab volunteer · R reporting ecological meta-analysis           | 2020      |
| <b>**Ashley Chen</b><br>Hanner Lab volunteer · R reporting ecological meta-analysis            | 2020      |
| <b>**Olivia Friesen Kroeker</b><br>Hanner Lab volunteer · R reporting ecological meta-analysis | 2020      |
| <b>*Steven French</b><br>CIS*4900/4910 · HACSim R package                                      | 2018      |
| <b>**Julia Harvie</b><br>MCB*4500/4510 · Data mining GenBank and BOLD                          | 2018–2019 |

### Graduate

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|---------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>**Nathan Zeinstra</b><br>IBIO*6070 · Bayesian habitat occupancy modelling for sea lamprey detection using eDNA               | 2024      |
| <b>**Amina Asif</b><br>BINF*6999 · DNA barcode gap analysis of Canadian disease vectors and agricultural pests                  | 2022      |
| <b>**Danielle St. Jean (withdrawn)</b><br>MSc. thesis (Mathematics) · DNA barcode sequence classification with machine learning | 2018–2019 |
| <b>**Christina Fragerl</b><br>BINF*6999 · DNA barcode sequence classification with machine learning                             | 2018–2019 |
| <b>**Jiaojia (Paula) Yu</b><br>BINF*6999 · MDMAPR R Shiny app                                                                   | 2018–2019 |
| <b>**Ankita Bhanderi</b><br>BINF*6999 · Data mining GenBank and BOLD                                                            | 2018      |

## TEACHING EXPERIENCE

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Course Instructor</b><br>University of Guelph<br><b>CIS*1910 – Discrete Structures in Computing I</b>                                                                                                                         | 2025                                   |
| <b>Graduate Teaching Assistant (GTA)</b><br>University of Guelph<br><b>CIS*3130 – System Modelling and Simulation</b><br><b>CIS*1910 – Discrete Structures in Computing I</b><br><b>CIS*2460 – Modelling of Computer Systems</b> | 2016–2020<br>2020<br>2017<br>2016–2019 |

## ACADEMIC SERVICE

|                                                                                                                                                                  |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6. <b>Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference</b><br>University of Guelph<br>• Conference organizer and volunteer       | 2026 |
| 5. <b>39<sup>th</sup> Canadian Artificial Intelligence Conference</b><br>Simon Fraser University<br>• Program committee reviewer for Long and Short Papers Track | 2026 |
| 4. <b>GBADs Informatics Technical Workshop</b><br>University of Liverpool<br>• Workshop organizer                                                                | 2023 |

3. **Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference** 2023  
University of Guelph  
• Conference organizer and volunteer
2. **School of Computer Science (SoCS) Faculty Search Committee** 2018  
University of Guelph  
• Member of hiring panel for Associate Professor in Cybersecurity
1. **School of Computer Science (SoCS) Faculty Search Committee** 2017–2018  
University of Guelph  
• Member of hiring panel for 2-year Contractually Limited Assistant Professor in Cybersecurity

## **CONFERENCE PRESENTATION AND PROCEEDINGS**

\* Indicates students under my direct mentorship/supervision

16. **The VII AMMCS International Conference** (student oral presentation) 2026  
Wilfrid Laurier University, Canada  
\*Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** · Modelling the Interplay between IUU Fishing, Wholesaler Fraud, and Consumer Awareness in the Seafood Supply Chain
15. **Pathway to Increase Standards and Competency of eDNA Surveys (PISCeS) International Conference** (oral presentation) 2026  
University of Guelph, Canada  
**Phillips, J.D.** · Association rule mining for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling
14. **Pathway to Increase Standards and Competency of eDNA Surveys (PISCeS) International Conference** (student poster presentation) 2026  
University of Guelph, Canada  
\*Toth, N., Antonie, M.L. and **Phillips, J.D.** · To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets
13. **39<sup>th</sup> Canadian Conference on Artificial Intelligence** (short paper and student poster presentation) 2026  
Simon Fraser University, Canada  
\*Toth, N., Antonie, M.L. and **Phillips, J.D.** · To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets.
12. **SoCS Undergraduate Summer Project Show and Tell** (student oral presentation) 2025  
University of Guelph, Canada  
\*Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** · The Importance in Design as a Computer Scientist
11. **CEPS Undergraduate Student Poster Day** (student poster presentation) 2024  
University of Guelph, Canada  
\*Toth N. and **Phillips, J.D.** · Association Rule Mining of eDNA Datasets
10. **CBS Undergraduate Poster Session** (student poster presentation) 2024  
University of Guelph, Canada  
\*Toth N. and **Phillips, J.D.** · Association Rule Mining of eDNA Datasets
9. **9<sup>th</sup> International Barcode of Life Conference** (poster presentation) 2024  
Estação das Docas, Belem, Brazil  
**Phillips, J.D.** · A Measure of the DNA Barcode Gap for Applied and Basic Research (*abstract accepted, not attended*)

8. **GBADs Informatics Technical Workshop** (oral presentation) 2023  
University of Liverpool, England  
**Phillips, J.D.** · The GBADs R Package (and Why We Need It!)
7. **CEPS Student Research Day** (student poster presentation) 2023  
University of Guelph, Canada  
\*De Vuono-Fraser, F.A. and **Phillips, J.D.** · Estimating Seafood Mislabelling Rates in Canada Using Bayesian Modelling
6. **Pathway to Increase Standards and Competency of eDNA Surveys (PISCeS) International Conference** (poster presentation) 2023  
University of Guelph, Canada  
**Phillips, J.D.** · Haplotype diversity reveals challenges and opportunities for developing targeted detection assays for *COI* in Canadian freshwater fishes
5. **8<sup>th</sup> International Barcode of Life Conference** (oral presentation) 2019  
NTNU University Museum and Norwegian Biodiversity Information Centre, Norway  
**Phillips, J.D.** · HACSim: Iterative extrapolation of haplotype accumulation curves for assessment of intraspecific COI DNA barcode sampling completeness
4. **Guelph BioMathematics and Statistics (BioM&S) Symposium** 2019  
**Artificial Intelligence and Machine Learning in Biology** (attended)  
University of Guelph, Canada
3. **CEPS Undergraduate Student Poster Day** (student poster presentation) 2018  
University of Guelph, Canada  
\*French, S.H. and **Phillips, J.D.** · Estimating Sampling Size Using Haplotype Accumulation Curves and Semiparametric Models
2. **7<sup>th</sup> International Barcode of Life Conference** (oral presentation) 2017  
University of Johannesburg, South Africa  
**Phillips, J.D.** · Intraspecific sample size estimation for DNA barcoding: Are current sampling levels enough?
1. **6<sup>th</sup> International Barcode of Life Conference** (poster presentation) 2015  
University of Guelph, Canada  
**Phillips, J.D.** · An exploration of sufficient sampling effort to describe intraspecific DNA barcode haplotype diversity: examples from the ray-finned fishes (Chordata: Actinopterygii).

## **REFEREED WORK**

**Publication record:** 9 peer-reviewed journal articles (7 first-author, 2 co-author) and 2 book chapters (1 first-author, 1 co-author), plus 3 first-author manuscripts in preprint or to be submitted. Senior (last) author on 2 student-led manuscripts

\* Indicates students under my direct mentorship/supervision

\*\* Indicates students under my co-mentorship/co-supervision

### **Journal Articles**

#### Published or Accepted

9. **Phillips, J.D.** and \*De Vuono-Fraser, F.A. (2026). Swimming in deep uncertainty: How proper statistical modelling can help expose seafood product mislabeling. *CHANCE*, 3. DOI: [10.1080/09332480.2026.2725526](https://doi.org/10.1080/09332480.2026.2725526)

8. **Phillips, J.D.** and \*De Vuono-Fraser, F.A. (2026). Statistical modelling of seafood fraud highlights uncertainties in products from Metro Vancouver, British Columbia, Canada: Revisiting Hu *et al.* (2018). *Journal of Food Science*, 91: e71201. DOI: [10.1111/1750-3841.71201](https://doi.org/10.1111/1750-3841.71201)
7. Raymond, K., Sobkowich, K.E., **Phillips, J.D.**, Nguyen, L., McKechnie, I., Mohideen, R.N., Fitzjohn, W., Szurkowski, M., Davidson, J., Rushton, J., Stacey, D.A., and Bernardo T.M. (2024). GBADs informatics strategy: User-centric tools, data quality, and model interoperability. *WOAH Scientific and Technical Review*, 43: 96–107. DOI: [10.20506/rst.43.3522](https://doi.org/10.20506/rst.43.3522).
6. **Phillips, J.D.**, Athey, T.B.T., Hanner, R.H., and McNicholas, P.D. (2023). VLF: An R package for the analysis of very low frequency variants in DNA sequences. *Biodiversity Data Journal*, e96480. DOI: [10.3897/BDJ.11.e96480](https://doi.org/10.3897/BDJ.11.e96480).
5. **Phillips, J.D.**, Gillis, D.J., and Hanner, R.H. (2022). Lack of statistical rigor in DNA barcoding likely invalidates the presence of a true species' barcode gap. *Frontiers in Ecology and Evolution*, 10: 859099. DOI: [10.3389/fevo.2022.859099](https://doi.org/10.3389/fevo.2022.859099).
4. D'Ercole, J., Dincă, V., Opler, P.A., Kondla, N.G., Schmidt, C.B., **Phillips, J.D.**, Robbins, R., Burns, J.M., Miller, S.E., Grishin, N., Zakharov, E.V., deWaard, J.R., Ratnasingham, S., and Hebert, P.D.N. (2021). A DNA barcode library for the butterflies of North America. *PeerJ*, 9: e11157. DOI: [10.7717/peerj.11157](https://doi.org/10.7717/peerj.11157).
3. **Phillips, J.D.**, \*French, S.H., Hanner, R.H., and Gillis, D.J. (2020). HACSim: An R package to estimate intraspecific sample sizes for genetic diversity assessment using haplotype accumulation curves. *PeerJ Computer Science*, 6(192): 1–37. DOI: [10.7717/peerj-cs.243](https://doi.org/10.7717/peerj-cs.243). *Top 5 most viewed article in category "Optimization Theory and Computation"*.
2. **Phillips, J.D.**, Gillis, D.J., and Hanner, R.H. (2019). Incomplete estimates of genetic diversity within species: Implications for DNA barcoding. *Ecology and Evolution*, 9(5): 2996–3010. DOI: [10.1002/ece3.4757](https://doi.org/10.1002/ece3.4757).
1. **Phillips, J.D.**, Gwiazdowski, R.A., Ashlock, D., and Hanner, R. (2015). An exploration of sufficient sampling effort to describe intraspecific DNA barcode haplotype diversity: examples from the ray-finned fishes (Chordata: Actinopterygii). *DNA Barcodes*, 3: 66–73. DOI: [10.1515/dna-2015-0008](https://doi.org/10.1515/dna-2015-0008).

#### Preprints

3. \*Toth, N., Antonie, L., Hanner, R.H., Gillis, D.J., and **Phillips, J.D.** Mining association rules for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling. bioRxiv. DOI: [10.64898/2026.09.16.752056](https://doi.org/10.64898/2026.09.16.752056).
2. **Phillips, J.D.**, Hubert, N., and Hanner, R.H. (2025). A Bayesian coalescent model of the DNA barcode gap. Authorea. DOI: [10.22541/au.174073683.37707806/v1](https://doi.org/10.22541/au.174073683.37707806/v1)
1. **Phillips, J.D.** and \*De Vuono-Fraser, F.A. (2024). Statistical modelling of seafood fraud in the Canadian supply chain. bioRxiv. DOI: [10.1101/2024.02.05.578947](https://doi.org/10.1101/2024.02.05.578947).

#### Submitted or Under Revision

1. \*Toth, N., Antonie, L., Hanner, R.H., Gillis, D.J., and **Phillips, J.D.** Mining association rules for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling. Submitted to *Oikos*.

#### To Be Submitted

1. **Phillips, J.D.**, Hubert, N., and Hanner, R.H. A Bayesian-informed coalescent model of the DNA barcode gap. Targeted to *Methods in Ecology and Evolution*.

### In Preparation (Selected)

3. \*Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** The influence of wholesaler fraud and consumer perception on IUU fishing in a model of the seafood supply chain. Targeted to *Proceedings of the National Academy of Sciences*.
2. Diao, J., Elliott, T.A., **Phillips, J.D.**, You, J., and Adamowicz, S.J. Benchmarking imputation methods for insect trait data across missingness mechanisms and trait types. Targeted to *Methods in Ecology and Evolution*.
1. D'Ercole, J., Dapporto, L., **Phillips, J.D.**, Dincă, V.E., Vila, R., Talavera, G., and Hebert, P.D.N. Macrogenetics of North American butterflies – The impact of Quaternary climatic fluctuations. Targeted to *Proceedings of the National Academy of Sciences*.

### **Book Chapters**

#### Published or Accepted

2. **Phillips, J.D.**, Griswold, C.K., Young, R.G., Hubert, N., and Hanner, R.H. (2024). A Measure of the DNA Barcode Gap for Applied and Basic Research. In: DeSalle, R. (eds) DNA Barcoding. Methods in Molecular Biology, vol 2744. Humana, New York, NY. URL: [https://link.springer.com/protocol/10.1007/978-1-0716-3581-0\\_24](https://link.springer.com/protocol/10.1007/978-1-0716-3581-0_24)
1. Hubert, N., **Phillips, J.D.**, and Hanner, R.H. (2024). Delimiting Species with Single-Locus DNA Sequences. In: DeSalle, R. (eds) DNA Barcoding. Methods in Molecular Biology, vol 2744. Humana, New York, NY. URL: [https://link.springer.com/protocol/10.1007/978-1-0716-3581-0\\_3](https://link.springer.com/protocol/10.1007/978-1-0716-3581-0_3)

#### Submitted or Under Revision

1. \*Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** Modelling the Interplay between IUU Fishing, Wholesaler Fraud, and Consumer Awareness in the Seafood Supply Chain. *VII AMMCS International Conference*, Springer Nature.

### **Conference Proceedings**

#### Published or Accepted

5. \*Toth, N., Antonie, M.L., and **Phillips, J.D.** To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets. *39<sup>th</sup> Canadian Conference on Artificial Intelligence*, **318**: 1012–1019. Short paper and poster.
4. Morey, K., Loeza-Quintana, T., **Phillips, J.**, and Hanner R. (2023). Haplotype diversity reveals challenges and opportunities for developing targeted detection assays for *COI* in Canadian freshwater fishes. Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference. Poster.
3. **Phillips, J.D.**, Gillis, D., and Hanner, R. (2019). HACSim: Iterative extrapolation of haplotype accumulation curves for assessment of intraspecific COI DNA barcode sampling completeness. Scientific abstracts from the 8th International Barcode of Life Conference, Trondheim, Norway (ed. Torbjørn Ekrem), *Genome*, 62(6): 349–453. Oral presentation.
2. **Phillips, J.D.**, Gillis, D., and Hanner, R. (2017). Intraspecific sample size estimation for DNA barcoding: Are current sampling levels enough? Scientific abstracts from the 7th International Barcode of Life Conference, Johannesburg, South Africa (ed. M. van der Bank), *Genome*, 60(11): 881–1019. Oral presentation.
1. **Phillips, J.D.**, Gwiazdowski, R.A., Ashlock, D., and Hanner, R. (2015). An exploration of sufficient sampling effort to describe intraspecific haplotype diversity in the ray-finned fishes (Chordata: Actinopterygii). Scientific abstracts from the 6th International Barcode of Life Conference, Guelph, ON., Canada (ed. S.J. Adamowicz), *Genome*, 58(5): 163–303. Poster.

## NON-REFEREED WORK

### *Ph.D. Thesis*

Phillips, J.D. (2022). A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling Completeness: Implications for DNA Barcode Gap Estimation. University of Guelph Atrium.

URL: <https://atrium.lib.uoguelph.ca/items/8addfcc5-f21c-4691-89b7-c4db051892eb>

\* Indicates students under my direct mentorship/supervision

14. \*Cui, R.C. and **Phillips, J.D.** (2026). Student-contributed poster for University of Guelph March Open House
13. \*Toth, N. and **Phillips, J.D.** (2026). Student-contributed poster for University of Guelph March Open House
12. \*Cui, R.C. and **Phillips, J.D.** (2025). The Importance in Design as a Computer Scientist. Student-contributed SoCS Show and Tell video URL: [https://github.com/RichaCui70/Dynamics-between-Seafood-Fraudsters-and-Buyers/tree/main/misc\\_resources](https://github.com/RichaCui70/Dynamics-between-Seafood-Fraudsters-and-Buyers/tree/main/misc_resources)
11. **Phillips, J.D.**, \*De Vuono-Fraser, F.A., Gillis, D.J., and Hanner, R.H. (2024). Statistical modelling of seafood fraud. Whiteboard explainer video. URL: <https://www.youtube.com/watch?v=4jU8cU8M70U>
10. \*Toth, N. and **Phillips, J.D.** (2024). eDNA Collection Gets a Tech Update. Student-contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2024/07/edna-collection-gets-tech-upgrade>
9. \*Toth, N. and **Phillips, J.D.** (2024). Unravelling eDNA with Association Rule Mining. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2024/08/15/unravelling-edna-with-association-rule-mining/>
8. **Phillips, J.D.** (2024). Summer URA Position. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2024/02/07/summer-ura-position-2/>
7. **Phillips, J.D.** (2022). Mind the Gap – The DNA Barcode Gap, That Is. Contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2022/08/mind-gap-%E2%80%93-dna-barcode-gap>
6. **Phillips, J.D.** (2020). Barcode Cracking. Contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2020/02/barcode-cracking>
5. **Phillips, J.D.** (2020). Protecting Biodiversity Through the Lens of Genetic Diversity. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2020/01/30/protecting-biodiversity-through-the-lens-of-genetic-diversity/>
4. **Phillips, J.D.** (2019). IBOL8 and the Midnight Sun. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2019/07/02/reflections-ibol8-and-the-midnight-sun/>
3. **Phillips, J.D.** (2017). The Big Five and IBOL7. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2017/12/06/reflections-the-big-five-and-ibol7/>
2. **Phillips, J.D.** (2016). Sample size estimation for DNA barcoding: Are current sampling levels enough? Contributed guest post to the DNA Barcoding Blog of Dr. Dirk Steinke. URL: <http://dna-barcoding.blogspot.com/2016/01/guest-post-sample-size-estimation-for.html>
1. **Phillips, J.D.** (2016). Sample size estimation for DNA barcoding of ray-finned fishes: Are current sampling levels enough? Contributed newsletter article to the Barcode Bulletin, 7(1). URL: <https://journal.lib.uoguelph.ca/index.php/ibol/issue/view/345>

## **ACADEMIC PEER REVIEW SERVICE**

*Ecology and Evolution, F1000 Research, Frontiers in Ecology and Evolution, Lifestyle Genomics, Mitochondrial DNA Part A, Molecular Ecology Resources, Molecular Biology Reports, Methods in Ecology and Evolution, Nature Communications*

## **VOLUNTEER EXPERIENCE**

### **CEPS Student Research Connections Networking Night**

2025

University of Guelph

- Connected with undergraduate students for summer URA project recruitment

### **CIS\*3750 Wireframing session**

2021–Present

University of Guelph

- Graded students on mobile app prototypes for various community partners using Qualtrics surveys